

Installing Software for Neuropixels analysis

Note: If you're viewing this in the github repo, the links won't be active. You can download the pdf to use the links (use the download button in the upper right corner).

This document describes setup for basic Neuropixels analysis, including KS4 and tools to build a simple pipeline. Sample data to test the installation is available through links in this github repository (<https://github.com/jcoutho/harvard-neuropixels-2026>). The goal of this exercise is for you to set up and run spike sorting. specifically on the test dataset. The repo also include Jupyter lab workbooks to extract sensory responses from the sort results.

For these demo data, a laptop with >16G RAM and a GPU with a Passmark rating > 10000 will be sufficient. If you are setting up an analysis workstation, there are guidelines for computer shopping here (https://github.com/billkarsh/SpikeGLX/blob/master/Markdown/SystemRequirements_PXI.md#spike-sorting-computer)

You'll want an ethernet connection (rather than wireless) to do the installation.s

If you are working on a system that already has sorting software installed, but want to learn how to do it yourself – go for it! Make sure that any virtual environments you create have new names.

These instructions are for the Windows OS. Most of the same instructions will work in Linux, but the testing was run in Windows.

Important Note #1 – If you don't have a computer with an appropriate GPU, you can still follow these instructions, simply downloading the results instead of running the pipeline.

Important Note #2 – If you already have your computer set up with your own tools (git, conda environments, python IDE, etc.), and just want a minimal install to run the tutorials in the repository, see the document: `Minimum_tutorial_software`.

Miniforge:

NOTE: If you already have Anaconda or Miniconda installed, feel free to use that instead. We encourage using Miniforge because there are no licensing fees for packages installed from its channels.

[GitHub - conda-forge/miniforge: A conda-forge distribution.](#)

Installer for windows included in 'software_downloads' folder

Default installation location

Do not add to environment path. (recent Windows 11 may skip this)

Register python as default for the system. (recent Windows 11 may skip this)

Remove package cache. (recent Windows 11 may skip this)

After installation, create an icon on the desktop

Git bash:

NOTE: We'll be accessing github through VSCode. If you already have any form of git installed, VSCode should work with that installation, and you should skip this step.

<https://git-scm.com/install/windows>

Download the installer. If a dialog is missing from this list, use the defaults.

- Accept license
- Default location
- Default components
- Default vim editor
- Switch from default to main for initial branch
- Default git from command line and 3rd party
- Default bundled SSH exe
- Default Windows Secure Channel Library
- Default line endings (windows checkout, unix check in)
- Default MinTTY for terminal
- Default git pull
- Default credential manager
- Default extra config options

VSCode:

NOTE: In our Neuropixels workshops we run sorting pipelines in VSCode, but they are compatible with any python IDE.

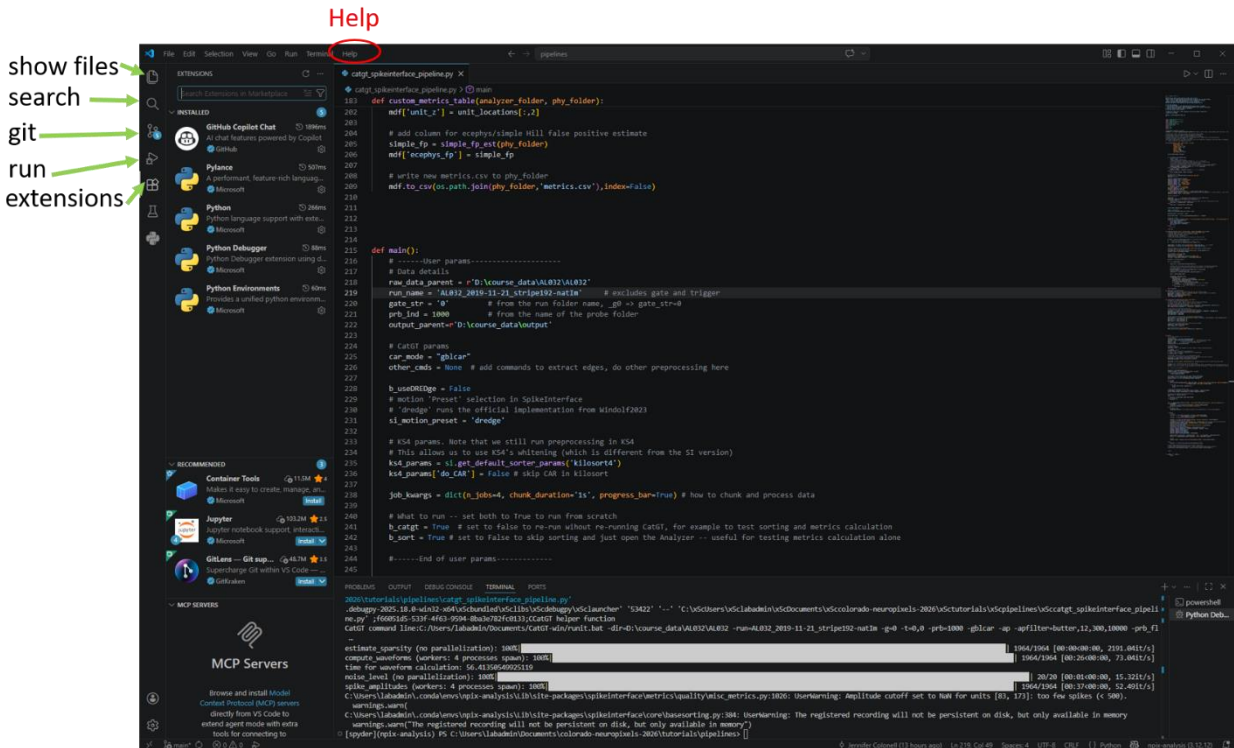
<https://code.visualstudio.com/download>

- accept license agreement
- default location OK

In “Additional Tasks” – if you want to minimize the impact of VS Code on your system – for example, if you use a different IDE and want it to be the default everywhere, uncheck all. Otherwise keep defaults.

A few useful features in the VSCode interface are shown in the screenshot below. There’s a great cheat sheet for VS code here: <https://www.learnbycheating.com/subjects/vscode/cheatsheet>

You can log in, or skip that (you can always log in later).



After opening VSCode, click on extensions, and install:

Python
 Pylance
 Python Debugger
 Python Environments
 Jupyter (this is a package of 4 extensions, install the group
 Jupyter Cell Tags
 Jupyter Keymap
 Jupyter Notebook Renderer
 Jupyter Slide Show

Some of these will come in groups (e.g. all the python stuff).

Clone the course repo:

Navigate to: <https://github.com/jenniferColonell/hands-on-neuropixels>

Click on the green <>Code button and copy the address for the http address for the repo

In VS Code, Help->Show all commands or ctrl-shift-P.

In the popup search for/select Git:Clone

Paste in the address. (Don't click 'Clone from github' – that will make you sign in)

VSCode will bring up a file dialog to choose the destination.

Select 'Open the repository' – this will make it a new VS code workspace.

To install environments:

NOTE: These will make new, unique environments, and won't step on any other software installations, as long as the environment names are different.

npix-test environment

From miniforge (or anaconda/miniconda) terminal:

```
conda create -n npix-test python=3.12 pandas natsort ipyml ipywidgets  
jupyterlab_widgets boto3 pyqt pyqtgraph jupyterlab zarr
```

You'll have to hit 'Y' for the installation to proceed.

Installing from conda helps resolve version issues, but handles DLLs in a way that can trigger security issues, especially with the latest Windows 11. One way to work around this is to install certain packages using pip. According to Gemini and Claude, pip installs these packages with the dlls incorporated into the compiled code, apparently avoiding the Smart App security system. Further internet wisdom: install the conda packages first, and run pip as a module inside the environment, to avoid installation with an unknown version:

```
python -m pip install scikit-learn scipy
```

Which DLLs get flagged by the Smart App security seems to be a moving target. If you see an error like "An Application Control policy has blocked this file", check which package is the problem, uninstall it with, for example (for numba):

```
python -m pip uninstall numba --force
```

```
python -m pip install numba
```

If you are skipping sorting, proceed to: Install phy2 environment

When that's complete, activate the environment.

```
conda activate npix-test  
python -m pip install kilosort[gui]  
python -m pip uninstall torch
```

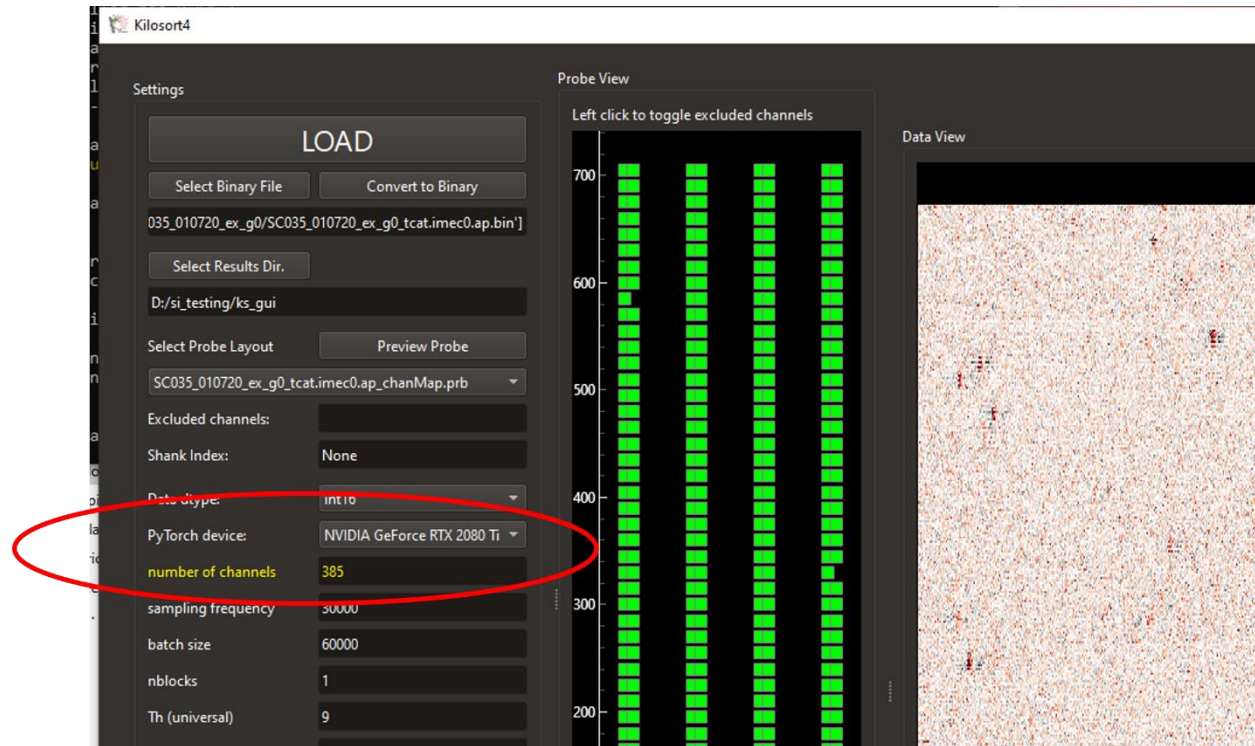
The last command removes the cpu version of torch that comes with the kilosort install. Next install pytorch with cuda. The cuda toolkit version is specified by the 'cuNNN' name at the very end. To figure out what cuda toolkit version you need, check the driver for your gpu, you need to know the driver version—go to the Device Manager->Display Adapters and get properties for your gpu. The Driver version is on the Driver tab of the Properties dialog, e.g. 31.0.15.3878. Just search for 'what cuda version is compatible with 31.0.15.3878'; for this driver, which is from 2024, the maximum supported version is 12.1. To find a version of torch that will be compatible, visit this website (<https://pytorch.org/get-started/previous-versions/>) and look through the history for a stable version that has a CUDA toolkit version as close as possible but not greater than the maximum supported. You can copy the command from torch website, or just edit the cu part of the command below:

```
python -m pip install torch torchvision torchaudio --index-url  
https://download.pytorch.org/whl/cu118
```

Test KS4 by running:

```
python -m kilosort
```

This can take a long time, in particular the first time it's run. The most common problem is that pytorch won't be able to "see" the GPU. The KS4 interface shows the active in the PyTorch device field. If it is not a GPU card, torch will need to be reinstalled.



This can occur due to incompatibility of the CUDA toolkit with the GPU or drivers; see above for (re) checking your drivers and installing.

The screenshot above shows what you'll see if a file is loaded in. The first time you load KS4

There's more information about debugging a Kilosort install in the github repo:

<https://github.com/MouseLand/Kilosort>

Finally, install spikeinterface:

```
python -m pip install spikeinterface
```

Install phy

Phy is a standard visualization environment for looking at sort results (<https://github.com/cortex-lab/phy>). The recommended install uses an entirely different virtual environment tool called uv. If your computer already has uv and phy installed, simply skip this step.

Installation instructions for uv can be found at: https://docs.astral.sh/uv/getting-started/installation/#_tabbed_1_2. For Windows, open a powershell window and use the command:

```
powershell -ExecutionPolicy ByPass -c "irm  
https://astral.sh/uv/install.ps1 | iex"
```

phy is then installed with:

```
uv tool install --python 3.12 phy
```

Test the installation with:

```
phy --version
```

SpikeGLX and CatGT installation (Windows)

The test data was acquired with SpikeGLX, so its viewer can be used to look at the data. The example pipelines use CatGT for preprocessing.

Make a new folder in a location where you have write access. For windows, download the standard SpikeGLX latest release and the windows version of CatGT, and simply unzip them in your new folder.

https://billkarsh.github.io/SpikeGLX/App/Release_v20251218-api4.zip

<https://billkarsh.github.io/SpikeGLX/Support/CatGTWinApp.zip>

SpikeGLX and CatGT installation (Linux)

https://billkarsh.github.io/SpikeGLX/Support/SpikeGLX_Linux_offline.zip

After unzipping, you may need to run:

```
chmod +x SpikeGLX-x86_64.AppImage
```

For CatGT, get the Linux version of the app:

<https://billkarsh.github.io/SpikeGLX/Support/CatGTLinuxApp.zip>

To install:

1. Copy CatGT-linux to your machine, cd into folder.
2. If needed, > chmod +x install.sh
3. > ./install.sh

Download the test data set

Create a folder for the raw data and output, on a drive with at least 100G of spare space. The folder name should not contain spaces.

The data download and later data visualization examples are presented in Jupyter lab notebooks. These can be opened in VSCode (more reliable) but should also work inside a browser (if you are using some other python IDE).

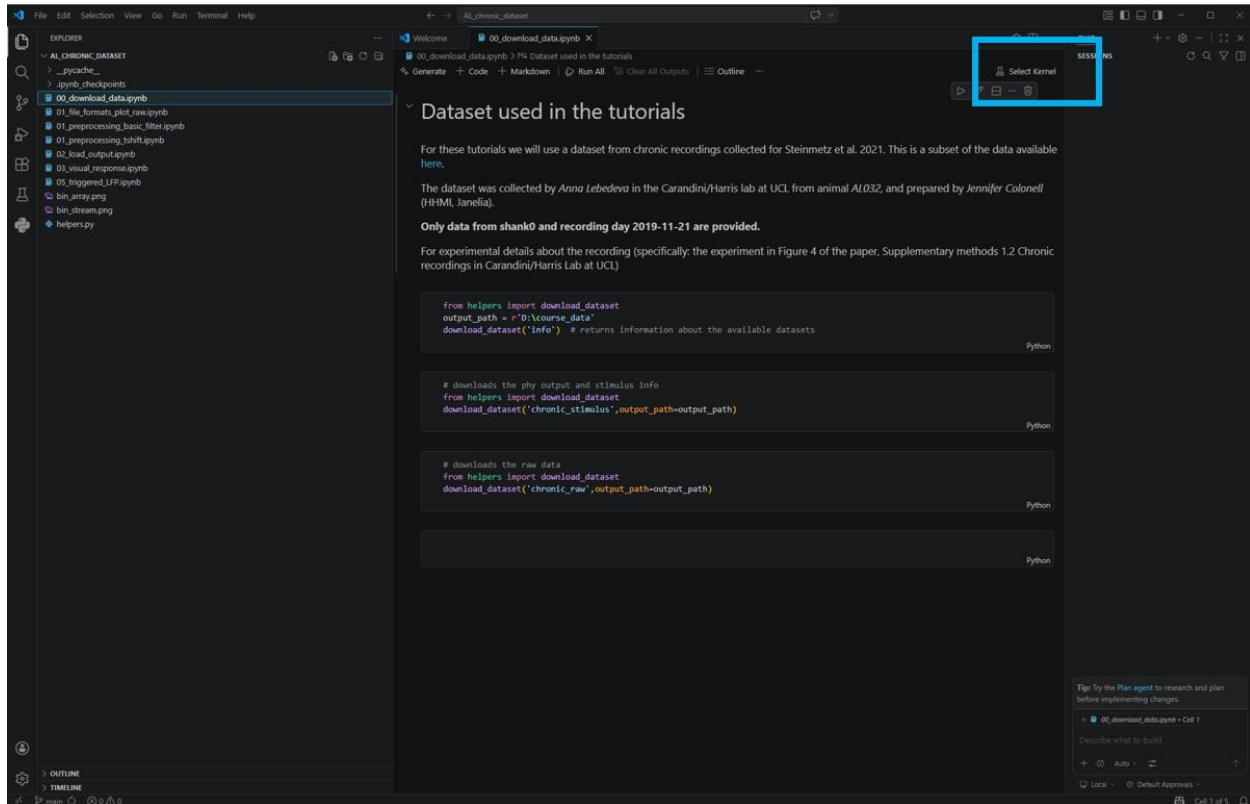
Opening the notebook in VSCode

File->Open folder to open:

hands-on-neuropixels/tutorials/AL_chronic_datasets

Click on 00_download_data.ipynb to open the notebook. In the upper right, click on “Select kernel” (see screenshot) and select the npix-test environment. If the environment isn’t showing up, cmd-shift-P to bring up the command palette, and ‘Set python interpreter’. Skip down to “Running the notebook” to proceed. If VSCode prompts you to set the folder as ‘Trusted’, do that, too.

Select kernel



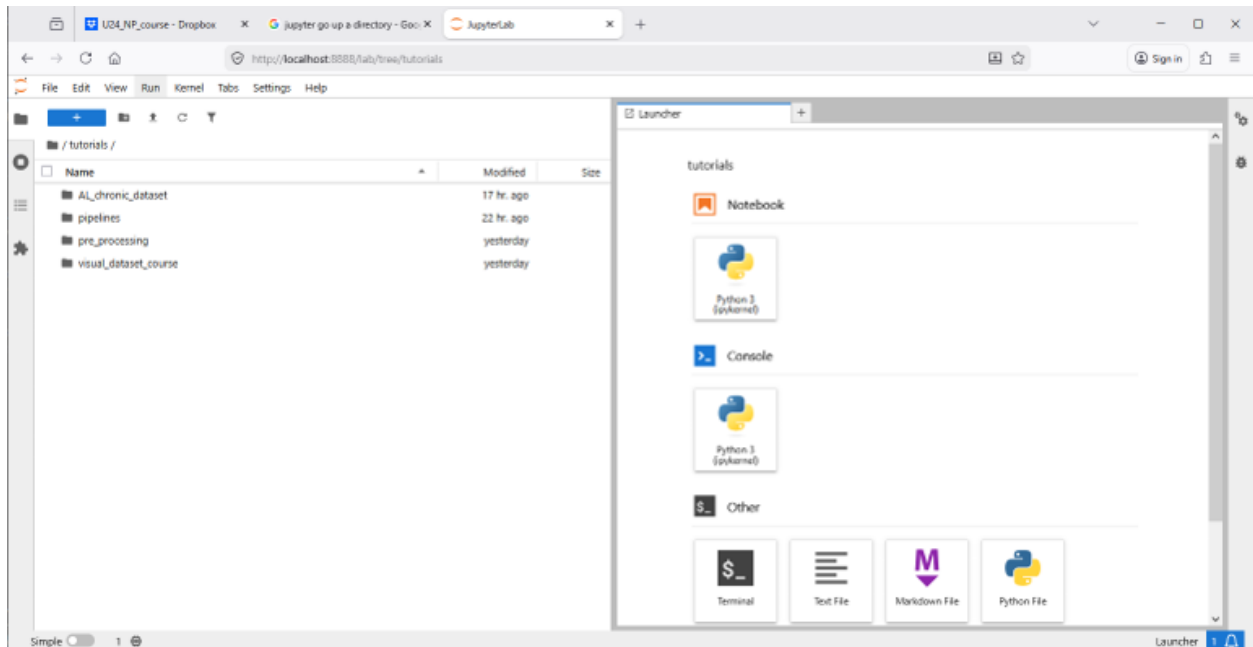
Opening the notebook in a browser

Open a miniforge (conda) terminal window and activate npix-test. Then navigate to the AL_chronic_dataset folder in the course repo, and start up jupyter lab:

```
conda activate npix-test
cd C:\Users\colonellj\Documents\hands-on-neuropixels
jupyter lab
```

jupyter lab will open in your default browser. If you run into any trouble with particular functions (especially interactive plots), it will probably work better in Chrome. The easiest way to force jupyter lab to open in Chrome is to set it to the default temporarily.

You'll get a screen like this, opened in the browser:



Double-click to open the folder `AI_chronic_dataset`, and then double click on `00_download_data.ipynb` to open a notebook to download the test data.

Running the notebook

Each block of code (cell) is run by hitting shift + return (you'll have to select the first one).

Before you run, change the variable `output_path` in the 2nd block to the output folder created above. If using windows paths, use the prefix `r` to avoid interpretation of the backslashes, e.g.:

```
output_path=r'D:\course_data'
```

Run each block of the code to download the raw data, stimulus times, and sample output to your computer. The raw data (in subfolder `AL032`) is in a zip archive, which must be extracted before running the pipeline.

Open the raw data in the SpikeGLX viewer

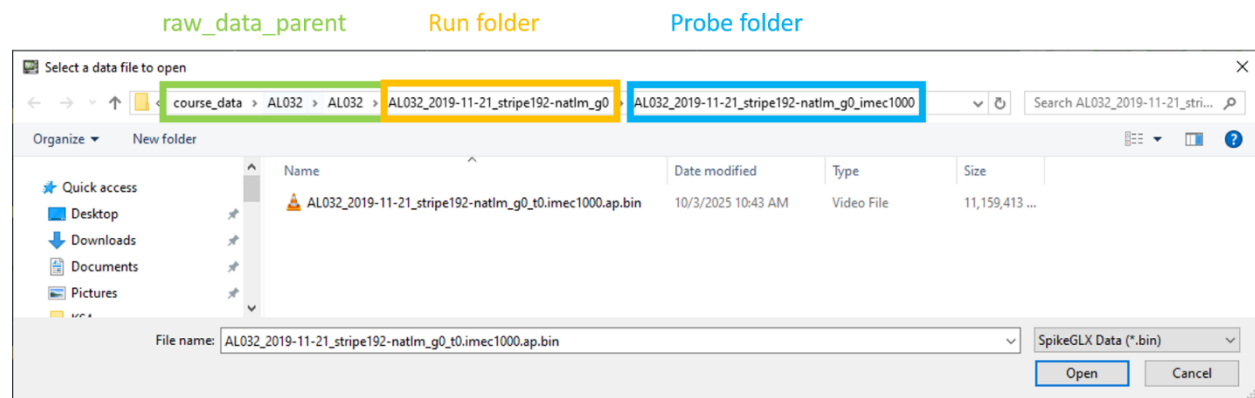
This data was collected using SpikeGLX. SpikeGLX data is in run folders, which contain a probe folder for each probe in the recording:

```
<run_name>_g<gate index>
  <run_name>_g<gate_index>_imec<probe index 0>
  <run_name>_g<gate_index>_imec<probe index 1>
  ....
```

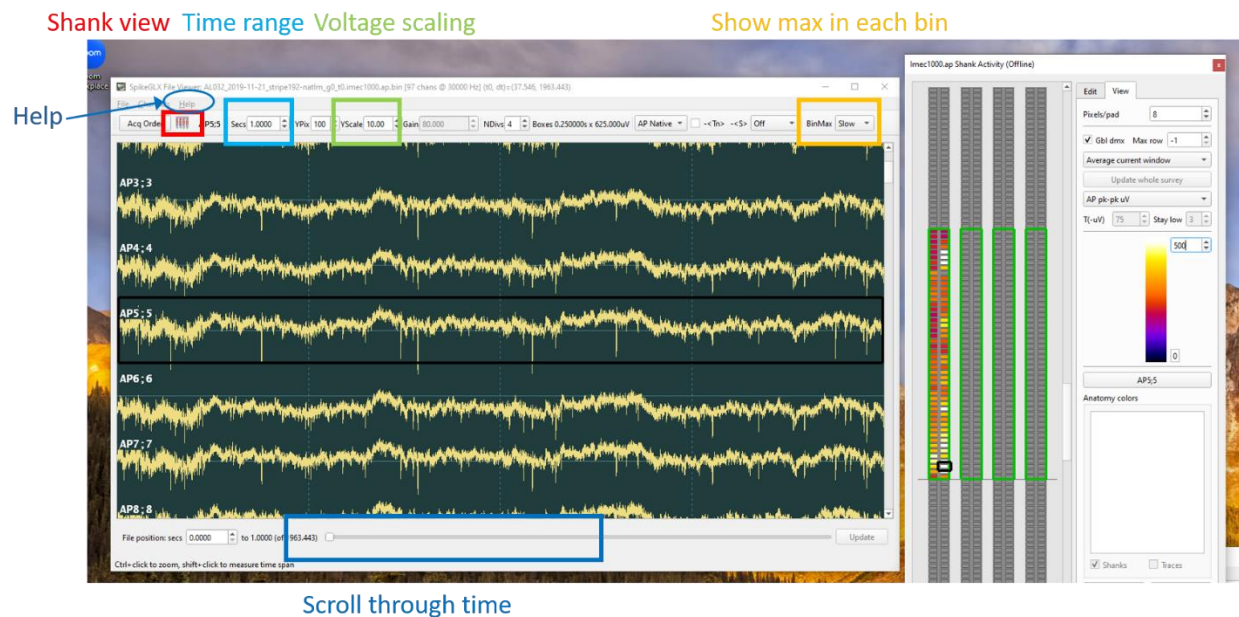
Each probe folder contains the binary raw data and a metadata file with all the required information to interpret the data: what sites were recorded, how many channels were saved, etc.

To view the data, run the `SpikeGLX_NSIM.exe` (in the folder: `Release_v20251218-api4\SpikeGLX`). Select: `File->Open File Viewer` and navigate into the probe folder, and select the binary. Here's what that

looks like in Windows:



The screenshot shows some useful scale settings. You can scroll in time with the bottom scroll bar, and see a summary of the activity by clicking on the shank view button. Click on Help for more options! After running the pipeline, the preprocessed data can be opened in the same viewer—a great way to confirm the preprocessing is having the intended effect.



Run the pipeline on the test data set

If you are skipping sorting, skip down to: Open the output in phy

We'll be running a simple pipeline to preprocess and sort this recording. It demonstrates how to call a couple of different tools (CatGT for preprocessing, SpikeInterface for sorting, calculation of some extra quality metrics). It's also run as a python script (.py) instead of a notebook (.ipynb) – this can be helpful for building a pipeline that will be run on many similar datasets.

To run .py file, you need to set the python interpreter (cmd-shift-P for command palette, then Python:Select Interpreter). It may also be necessary to set the default terminal window to Windows Command Prompt, rather than the default powershell. In the command palette, find Terminal: Select

Default Profile, and select 'Command Prompt'. Close any current terminal windows. To run the script, select Run->Start Debugging, and the default debugger.

Steps to run this pipeline:

Create folder for the sorting output, e.g.:

D:\course_data\output

Open VS code:

View->Explorer to show files in the left hand side bar

File->Open folder to navigate to hands-on-neuropixels/tutorials/pipelines folder

Double click to open catgt_spikeinterface_pipeline.py

Help->Show All Commands

In the command input, type in 'Python Select Interpreter' and select the npix-test environment.

To run the code:

Set CatGTPath to the path to the CatGT exe (line 33).

In the main() routine, set paths and other information about the data you want to sort e.g.

```
raw_data_parent = r'D:\course_data\AL032\AL032' # The directory that contains the run folder
run_name = 'AL032_2019-11-21_stripe192-natlm' # excludes the gate and trigger
gate_str='0'
probe_ind=1000
output_parent = r'D:\course_data\output'
```

The gate string is the part of the run folder name after the 'g', e.g. AL032_2019-11-21_stripe192-natlm_g0 is '0'. The probe index is included in the name of the probe folder and the binary, with the prefix 'imec'. 'imec1000' in the binary name indicates that the probe index is 1000. (Note: This was not a 1000 probe experiment—the index was chosen when extracting the data from shank 0 for this example.)

From here, you can simply run the code. (Run -> Start Debugging, the run button).

Most errors will be due to problems with the folder naming. In particular, if CatGT fails, check the CatGT.log to see if it seems to be looking for the wrong folder.

Steps during the run:

CatGT runs preprocessing. This doesn't report anything to the command line, but the VS code status bar will be blue, indicating its active. Depending on your hard drive, this will take 5-6 minutes.

Spike interface loads the preprocessed recording.

KS4, called from SpikeInterface – its output is streamed to the terminal, so you'll be able to see progress. Run times will be variable, but expect ~10 minutes. If KS4 is running very slowly, check the command line output for GPU memory usage in the 'Resource Usage' blocks. If it is 0, then KS4 is running on the CPU, and that will need to be debugged.

SpikeInterface builds Analyzer object and calculates metrics.

Pipeline code calculates a few more metrics.

Open the output in phy

Open a new conda window, and activate the phy2 environment, then cd to the directory of sorter output.

If you didn't run the sorter locally: To look at the sorter output that came with the download, navigate to that directory. The default is <your home directory>\course_data)\AL032_output\AL032_2019-11-21-natIm_g0\kilosort4_output\sorter_output

If you ran the pipeline and saved the output to the D drive, the commands are:

```
conda activate phy2-test
d:
cd course_data\output\AL032_2019-11-21_stripe192-
natIm_g0\kilosort4_output\sorter_output
```

Inside this folder there's a family of output files describing the sort output. A good overview of what's in these files (from Nick Steinmetz) is available here, along with advice about how to use phy on real data: https://phy.readthedocs.io/en/latest/sorting_user_guide/. The folder also contains the params.py file that specifies data details for phy.

We'll talk much more about what's displayed in the course, the main goal right now is to confirm the phy installation. Open the GUI with:

```
phy2 template-gui params.py
```

These warnings in the command line are all normal:

```
(phy2_new) D:\course_data\AL032_output\AL032_2019-11-21_stripe192-natIm_g0\kilosort4_output\sorter_output>phy template-gui params.py
17:37:56.956 [W] model:1416      File D:\course_data\output\AL032_2019-11-21_stripe192-natIm_g0\kilosort4_output\sorter_output\recording.dat does not exist.
17:37:57.263 [W] model:668      Skipping spike waveforms that do not exist, they will be extracted on the fly from the raw data as needed.
17:37:57.267 [W] traces:461     File D:\course_data\output\AL032_2019-11-21_stripe192-natIm_g0\kilosort4_output\sorter_output\recording.dat does not exist.
17:37:57.270 [W] traces:461     File D:\course_data\output\AL032_2019-11-21_stripe192-natIm_g0\kilosort4_output\sorter_output\recording.dat does not exist.
```

The interface will come up with no plots, click on a unit in the ClusterView (top left) to see attributes of that cluster.

The warnings occur because the params.py file is specifying the SpikeInterface internally generated recording.dat file as the raw data. For these data, the preprocessed output from CatGT is a good source for the raw waveforms and amplitudes. To use that binary, edit three fields in params.py:

- `n_channels_dat` needs to reflect the original data, which includes the sync/status word in addition to the 96 data channels. In general, these details can be found in the metadata file.
- `hp_filtered` needs to be set to True.
- `dat_path` needs to point to the CatGT output. Note the use of forward slashes, another way to avoid python interpreting backslashes in the path.

```
n_channels_dat = 97
offset = 0
sample_rate = 30000.0
dtype = 'int16'
hp_filtered = True
```

```
dat_path = ['D:/course_data/output/AL032_2019-11-21_stripe192-natIm_g0/  
AL032_2019-11-21_stripe192-natIm_g0_imec1000/AL032_2019-11-  
21_stripe192-natIm_g0_tcat.imec1000.ap.bin']
```

Reopening the data with this params.py file will allow display of waveforms from the data, which is useful for assessing jitter and signal to noise.